

Optimal Page Replacement Algorithm

1 Core Idea

Definition

The **optimal page replacement algorithm** evicts the page whose next use is farthest in the future.

When a page fault occurs, the operating system must choose one page already in memory as the victim. The optimal rule is simple: remove the page that will not be needed for the longest time.

Optimal Rule

evict the resident page with the farthest next reference

2 How the Decision Is Made

Imagine each page in memory has a label showing when it will be used next.

Page	Next use
<i>A</i>	After 10 instructions.
<i>B</i>	After 100 instructions.
<i>C</i>	After 1000 instructions.
<i>D</i>	After 8,000,000 instructions.

Decision

Evict page *D*, because it will be unused for the longest time.

3 Worked Example

Problem

Memory has 3 frames containing *A*, *B*, and *C*. A page fault occurs for *D*. Future references are:

B, *C*, *A*, *B*, *E*, *C*, *A*

Which page should be evicted?

Solution

Resident page	Next use in the future string
<i>A</i>	Third.
<i>B</i>	First.
<i>C</i>	Second.

Page *A* is used farthest in the future, so it is the best victim.

Answer

Evict *A*, then load *D* into the freed frame.

4 Why It Is Impossible in Practice

Main Limitation

The operating system does not know the future page-reference sequence when the page fault happens.

To apply the optimal rule, the OS would need to know exactly:

- which resident page will be used next;
- how long until each resident page is used again;
- whether any resident page will never be used again.

Real systems only know past and current behavior, not exact future references.

5 Why It Is Still Useful

Benchmark

Optimal replacement gives the lowest possible page-fault count for a known reference string.

Researchers can record a program's page references in a simulator, then replay that same reference string and compare real algorithms against the optimal result.

Use

It is not a practical operating-system algorithm. It is a standard for measuring how close practical algorithms are to the best possible result.

6 Final Summary

Point	Meaning
Rule	Evict the page whose next use is farthest away.
Strength	Gives the best possible page-fault performance.
Weakness	Requires future knowledge, so it cannot run in a real OS.
Main use	Benchmark for comparing practical replacement algorithms.

One-Box Summary

Optimal replacement is theoretically best.
It evicts the page needed farthest in the future.
It is useful for comparison, not implementation.